

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (EC)

May 2014

EC403 Mobile and Satellite Communication

Date: 21.05.2014, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Draw the timing diagram of call origin to call connects between Mobile and PSTN. [07]
- Q - 2 (a) Discuss various points on improving coverage and capacity in cellular systems. [07]
- (b) Calculate the mean path loss using okumura's model for $d=50\text{Km}$, $h_{te}=100\text{m}$, $h_{re}=10\text{m}$ in a suburban environment. If the base station transmitter radiates an EIRP of 1Kw at a carrier frequency of 1800 MHz, find EIRP (dbm) and the power at the receiver where gain at receiving antenna is 10dB. Assume median attenuation relative to free space is 48dB, Area due to type of environment is 25dB. [07]

OR

- Q - 2 (a) Write a note on following terms [07]
- a) Cochannel interference
- b) Adjacent channel interference
- (b) Which three basic propagation mechanisms impact in mobile cellular system? Explain any one of them. [07]
- Q - 3 (a) Describe how intersystem handoff mechanism works in mobile cellular system. [06]
- (b) Define following terms [08]
- i) Umbrella cell
- ii) Cell dragging
- iii) Hard handoff
- iv) Soft handoff

OR

- Q - 3 (a) Give different channel assignment strategies for mobile communication. And explain the same. [06]
- (b) Draw the chart of types of small scale fading and explain each one with necessary detail. [08]

SECTION – II

- Q - 4 (a) Explain Code division multiple access system in detail. [07]
- Q - 5 (a) Draw the schematic block diagram global system for mobile (GSM) system architecture and explain each block in detail. [07]
- (b) Describe minimum shift keying (MSK) transmitter and receiver with necessary diagram. [07]

OR

- Q - 5 (a) Explain the “mobile call termination” for Global System for Mobile Communication. [07]
- (b) Explain equalization, diversity and channel coding for wireless communication system. [07]
- Q - 6 (a) Define [07]
- i) Ascending node
 - ii) Descending node
 - iii) Mean anomaly
 - iv) True anomaly
 - v) Apogee
 - vi) Perigee
 - vii) Inclination
- (b) Comparisons are sometimes made between satellite and optical fiber communication systems. State briefly the area of application for which you feel each system is best suited. [07]

OR

- Q - 6 (a) State equivalent isotropic radiated power (EIRP). Calculate the gain of a 3Meter parabolic reflector antenna at a frequency of [07]
- i) 6 GHz
 - ii) 14GHz.
- Assume an aperture efficiency of 0.60.
- (b) Explain what is meant by carrier to noise ratio. A satellite TV signal occupies the full transponder bandwidth of 36 MHz, and it must provide a C/N ratio at the destination earth station of 22 dB. Given that the total transmission losses are 200 dB and the destination earth station G/T ratio is 31 dB/K, calculate the satellite EIRP required. [07]
